Enterprise Multi-Site Network Design & Implementation

Cisco Packet Tracer Architecture, Routing, & Security Implementation

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Executive Summary

This project presents a scalable, multi-site enterprise network infrastructure connecting a Headquarters (HQ) site with a Branch Office using Cisco ISR4331 routers and Catalyst 2960 switches. The design enforces department isolation via IEEE 802.1Q VLAN trunking, enables automated IP addressing via dynamic DHCP pools, establishes inter-site reachability through single-area OSPF (Area 0), and secures critical management infrastructure using Extended Access Control Lists (ACLs).

1. VLAN & Subnet Addressing Schema

Location	Department	VLAN Name	VLAN ID	Device / Host	Subnet / Mask	Switch Port	Gateway (Router Port)
HQ Site	IT Dept	IT_Dept	10	PC0	192.168.10.0/24	Fa0/2	Gig0/0.10 (192.168.10.1)
HQ Site	HR Dept	HR_Dept	20	PC1	192.168.20.0/24	Fa0/10	Gig0/0.20 (192.168.20.1)
HQ Site	Management	Management	99	Server0 (Web/Mgmt)	192.168.99.0/24	Fa0/5	Gig0/0.99 (192.168.99.1)
Branch Office	Sales Dept	Sales_Dept	30	PC2	192.168.30.0/24	Fa0/2	Gig0/0.30 (192.168.30.1)

2. Inter-Switch & WAN Link Configuration

Link Type	Source Device	Source Port	Destination Device	Destination Port	Protocol / Configuration
Trunk Link (HQ)	SW1-HQ	Fa0/1	R1-HQ	Gig0/0	IEEE 802.1Q Trunking (VLANs 10, 20, 99)
Trunk Link (Branch)	SW2-Branch	Fa0/1	R2-Branch	Gig0/0	IEEE 802.1Q Trunking (VLAN 30)
WAN Interconnect	R1-HQ	Gig0/1	R2-Branch	Gig0/1	Point-to-Point 10.0.0.0/30 (OSPF Area 0)

3. Extended ACL Security Implementation & Verification

Access Control Requirement: Restrict Branch Office (VLAN 30 - Sales) from accessing the HQ Management Server (192.168.99.10), while maintaining full inter-department routing for authorized traffic.

ACL Logic & Enforcement Point: Applied extended ACL BLOCK_SALES_SERVER inbound on R1-HQ interface GigabitEthernet0/1.

- ip access-list extended BLOCK_SALES_SERVER
- deny ip 192.168.30.0 0.0.0.255 host 192.168.99.10
- permit ip any any

Verification Results:

1. **PC0 (HQ IT) → Server0 (192.168.99.10):** ICMP Ping Successful (100% Reachability).
2. **PC2 (Branch Sales) → PC0 (192.168.10.2):** ICMP Ping Successful across OSPF Area 0.
3. **PC2 (Branch Sales) → Server0 (192.168.99.10):** Blocked by ACL (*Destination host unreachable*). Security enforced.